

New Microprocessor Based Relay to Monitor and Protect Power Systems Against Sub-Harmonics

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Abstract-- A routine switching event in the Xcel Energy transmission system resulted in sustained oscillations between wind generators and the system connected through a series compensated line. This event led to the development of a new microprocessor based sub-harmonic protection relay that could detect such conditions and take preventative measures.

This paper discusses a new technique to detect sub-harmonics in the range of 5 - 25 Hz using current and voltage level detectors, sub-harmonic calculations of individual sub-harmonics and total sub-harmonic distortion factor (TSHD) to generate an alarm and/or trip to isolate the faulty system. Various test cases are presented to illustrate the effectiveness of the new technique.

Key words-- Series-compensated wind farms, transmission line protection, sub-harmonic oscillations, sub-harmonic protection, Recursive Fourier Transform

I. INTRODUCTION

THE goal of a power system is to provide reliable power from generation sources to customer loads. In the process of providing this power, the power system can be affected by the interaction of various components that can excite non-characteristic sub-harmonic frequencies (below 60 Hz or 50 Hz power system frequency) [1]. This effect is particularly prevalent on the power system where wind farms, typically located far away from load centers, are interconnected with the rest of the power grid via series compensated lines [2-4]. The wind turbine mechanical system interactions (tower-to-blade) can also generate sub-harmonics, which are detrimental to induction generators and transformers, and may cause resonance at the point of common coupling (PCC) in the electrical grid [5].

Two sub-synchronous oscillation effects have been described in the literature – sub-synchronous oscillations resulting in exciting the mechanical modes of oscillations on

the turbine generator shaft and the induction generator effect on wind generators.

Sustained sub-synchronous oscillations on wind machines due to low damping coefficient are analyzed and solution methods are presented in the following sections.

The paper is organized as follows: in section II, the Xcel system where this problem occurred is described. In section III, the proposed solution is discussed and details of the relay hardware architecture, the sub-harmonic detection algorithm and the monitoring process are described. This is followed by the tests and results in section IV and the discussion and conclusions are provided next.

II. SYSTEM

Installations of several hundred MWs of wind generation during 2004-2008 in the Buffalo Ridge area of Minnesota resulted in significant improvements in transmission system in Minnesota and South Dakota. In order to limit the additional loading of constrained interfaces and to direct power flow towards Twin Cities, Xcel Energy installed series compensation on a 54 mile long 345 kV line in southwestern Minnesota. The degree of compensation was decided to be 60% based on planning studies and with the use of parametric analysis to determine the effect on constrained interface loading. A 240 MVAR series capacitor was installed in the middle of the line. One end of this series compensated line also has 150 MW of installed wind generation and six combustion turbine generation (CTG) totaling 639 MW as shown in Figure 1.

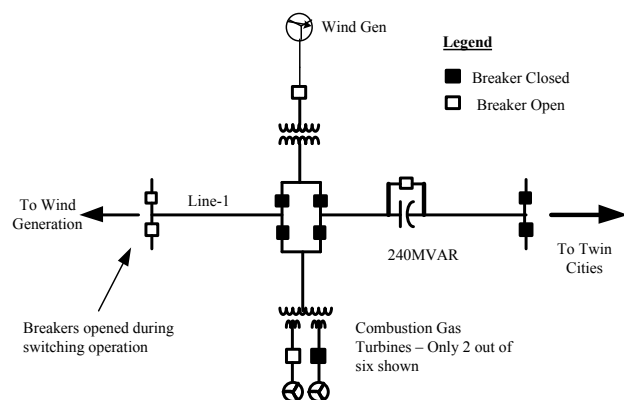


Fig.1 System one line showing generation and series compensated line.

During the commissioning process of series capacitor and associated controls, Line -1 was taken out of service before

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bypassing the capacitor. At the time of the event, the total wind generation was 15 MW and one of the six CTG units was generating 46 MW. All other units were not on line. This switching operation resulted in radial connection of generation to the system through the series compensated line. The system tripped due to a flashover in the generator bus duct resulting in a bypass of the series capacitor bank. The oscillation started by the switching event continued in the wind generators even after the series capacitor was bypassed.

The generator current waveforms captured are as shown in Figure 2. Analyses of the captured waveforms indicated the presence of DC and low frequencies between 9-13 Hz indicating different oscillatory interactions between wind machines and the system. These frequencies were below the sub-synchronous resonance mode 1 frequency of the CTGs.

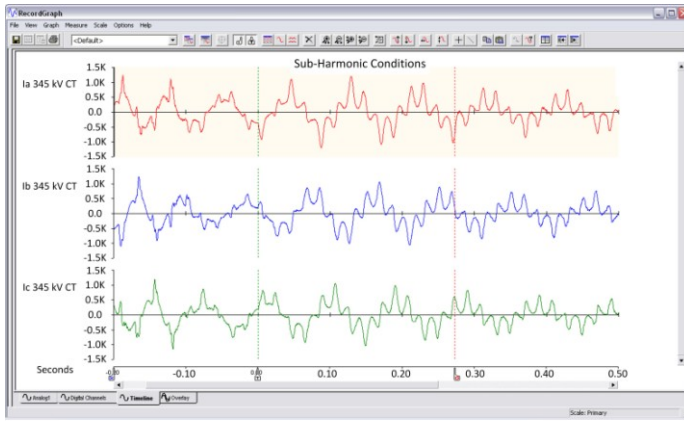


Fig. 2 Sub-harmonic oscillations at wind farm interconnections.

III. SOLUTION

Analysis of Xcel's system in relation to their experiences as described above, led to the realization that they needed accurate detection of sub-harmonic oscillations on their system and protection schemes to isolate disturbances when they occur. By isolating the healthy grid from sub-harmonic generation sources, damage to the electrical interconnection could be prevented.

A new algorithm for a protection relay to detect sub-harmonic frequencies was therefore needed since existing relays could not perform this function.

The first generation of a new sub-harmonic protection relay is now in service in Trimont, MN, USA, detecting sub-harmonic oscillations occurring at adjacent wind farms connected to the series compensated line. It monitors system voltages and currents on the power system that may be present in the 5 - 25 Hz frequency window. It can be programmed to either detect/record only the sub-harmonic quantities for future use in simulations, or to initiate protective action. In tests so far, the unit has performed well and the relay has been designed to detect and protect and protect against sustained harmonics from any source.

A. Micro processor based hardware architecture

The microprocessor based device can accept four sets of 3-phase currents and two sets of 3-phase voltages from instrument transformers. Two sets of level detectors can be set for each input quantity to detect specific frequency or all components in the 5-25 Hz range. The device also has the ability to sum quantities from two current transformers (CTs) on a ring or breaker-and-a-half bus arrangement that might represent the total current on a particular line, and then apply the level detectors to these summated quantities. The device also has triggered-recording capability to alarm or take action for cyclical sub-harmonic quantities that may be oscillating just below the tripping levels.

The intention here is to collect sub-harmonic data as it occurs even if it is on the verge of being a problem. The pickup levels and time delays of the detectors, and resulting action, is fully user selectable. The device has the ability to collect and store fault and harmonic voltage and current quantities. The device samples the input signals at 96 samples per cycle (60 Hz or 50 Hz system), so with its anti-aliasing filtering, up to the 25th harmonic is available. Following this process, a complete range of power system data is collected for further analyses by the user. All detector and event operations are time tagged and presented in an event log. IIRIG-B time synchronization is possible by connecting this device to a GPS clock. All information and data collected can be retrieved remotely via the local area network (LAN) through Ethernet ports, or locally via the USB or Ethernet port located on the front of the device. Figure 3 shows the block diagram of the microprocessor based hardware architecture used to detect the sub-harmonic oscillations.

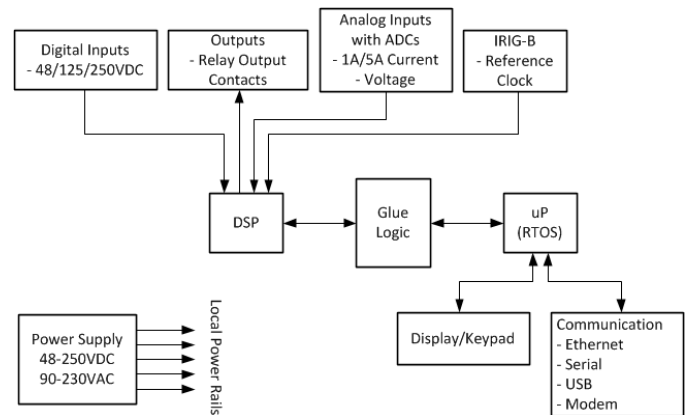


Fig. 3 - Hardware Architecture Block Diagram

B. Sub-Harmonic Detection Process

The analog signals from the CTs and PTs are fed to the low-pass analog filter to eliminate aliasing effects, and then passed to the A/D convertor where the signals are converted to digital samples. The digital samples of each analog channel and the derived virtual channels are evaluated for the sub-harmonic frequencies from 5-25 Hz with a resolution of 1 Hz, using Recursive Fourier Transform (RFT). The extracted sub-harmonic signals pass onto the sub-harmonic detection logic

module for comparison, and based on the detector level, the output contacts will be actuated for alarm and/or trip conditions. The simplified process is illustrated in Figure 4.

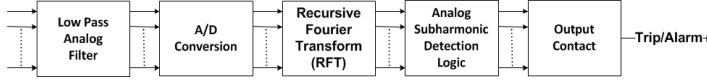


Fig. 4 Sub-harmonic Detection Process Logic

C. Principle of sub-harmonic detection

The basic principle used in detecting the sub-harmonic is to compare the magnitude of each sub-harmonic between the minimum F_{min} and maximum F_{max} frequencies of the user defined frequency range, and then compare it with the user defined (L_{set}) magnitude threshold level using the following logic (Figure 5):

$$\text{Trip or Alarm} = \max(f_2, f_3, f_4, f_5, f_6, f_7) > L_{set} \quad (1)$$

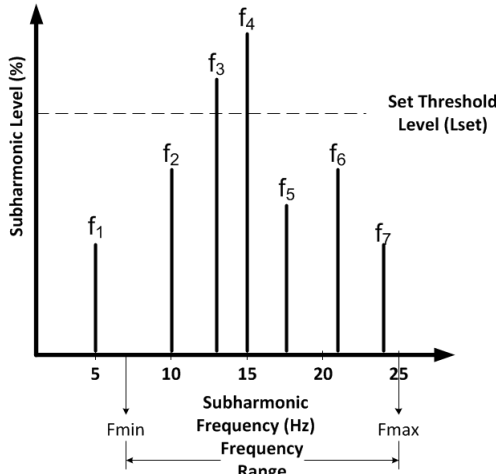


Fig. 5 Sub-harmonic Detection Principle

The functional logic diagram corresponding to the above principle is illustrated in Figure 6. As described in the figure, there are three level detectors available for each voltage, current, or virtual input.

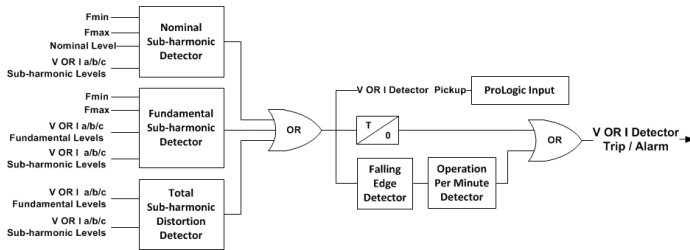


Fig. 6 Sub-harmonic Detection Logic Diagram

The nominal sub-harmonic detector compares the level by estimating the ratio of sub-harmonic magnitude to the nominal CT/PT ratio. For example, if the nominal CT ratio is 5A and the estimated sub-harmonic magnitude is 2.5 A at 5 Hz, then the nominal sub-harmonic level is compared as follows:

$$f1_{5Hz} (\%) = (2.5/5) * 100 \quad (2)$$

Similarly, the fundamental sub-harmonic detector logic is compared with respect to the fundamental voltage or current quantities.

The Total Sub-Harmonic Distortion (TSHD) detector calculates the distortion level as follows:

$$TSHD(\%) = \left(\frac{\sqrt{f_{5Hz}^2 + f_{6Hz}^2 + f_{7Hz}^2 + \dots + f_{25Hz}^2}}{f_{60Hz}} \right) 100 \quad (3)$$

Note that, as shown in the above equation, all the sub-harmonic magnitudes from 5-25 Hz will be taken into consideration for the TSHD evaluation, with respect to 60 Hz fundamental voltage, current, or virtual derived channel. The same definition is applicable for a 50 Hz system.

D. Operations / Minutes Detector

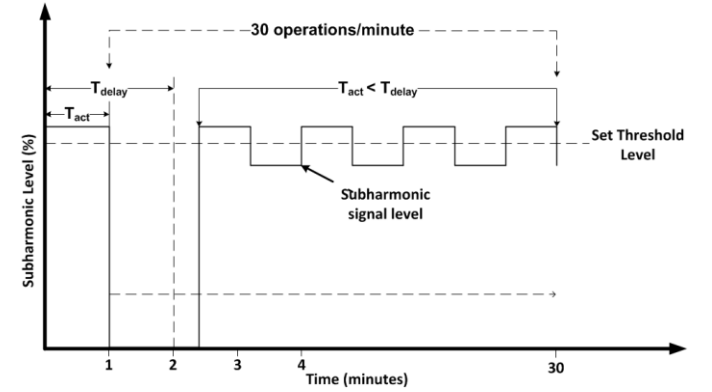


Fig. 7 Operations / Minute Detector Functionality

There are occasions when the statistical nature of the wind generation and its interactions with the rest of the system may produce sub-harmonic oscillations above threshold set limit for a lesser time period than the configured time delay, and will never be noticeable by the conventional detectors mentioned in the above section. Periodic occurrence of this event, even though for a shorter duration than the configured time, can have a negative impact in the power system network and its components. To capture such events, a special operations/minutes detector is designed, which functions as shown in Figure 7.

In the above example, an event with 30 operations per minute is depicted (not to scale). The time T_{act} corresponds to the actual duration of the sub-harmonic signal which is asserted. This event is not captured by the conventional detectors mentioned in the previous section, as the pickup delay T_{delay} has not been exceeded so the event is not noticeable. However, with the special detector, the 30 operations (assertion above set limit) will be internally counted and monitored. If the set operations per minute count exceed the calculated count, then this special detector will issue a trip or an alarm as per the configuration. In this way, periodic

disturbances with durations shorter than the configured limit can be captured.

IV. TESTS AND RESULTS

Several test cases were carried out to verify the sub-harmonic protection technique using a Real Time Digital Simulator (RTDS) to test the performance of the following relay functions:

- 1. Nominal sub-harmonic detector
- 2. Fundamental sub-harmonic detector
- 3. Total sub-harmonic detector
- 4. Operations/Minute detector

The settings screen shown in Figure 8 is used to set the various detector levels. As explained in the previous section, each current, voltage, or virtual derived channel can be configured to have two detectors, and each can be used for an alarm and/or trip output.



Fig. 8. Settings screen for configuring various sub-harmonic detector levels.

The real-time waveforms required to test the relay were generated from a real time digital simulator (RTDS). The schematic diagram of test setup used for relay testing is shown in Figure 9. The analog waveforms generated from RTDS were sent to the relay via an amplification circuit. The trip signal generated by the relay was fed back into the RTDS. The user interface for RTDS and relay was provided using a personal computer.

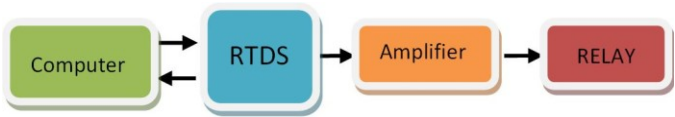


Fig. 9: Test setup

As shown in Figures 10a-10d, the nominal, the fundamental, the TSHD, and the operations per minute detectors have been verified using simulated COMTRADE files. These files were then played back using the RTDS simulator, and the detectors have operated correctly as per their configurations.

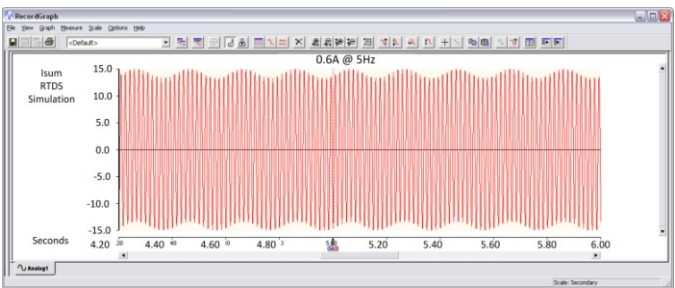


Fig. 10(a): Nominal sub-harmonic detector – 5 Hz (X axis – time in seconds), (Y axis – current in amps).

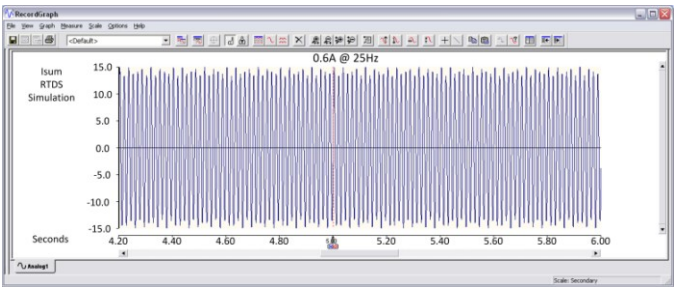


Fig. 10(b): Fundamental sub-harmonic detector – 25 Hz (X axis – time in seconds), (Y axis – current in amps)

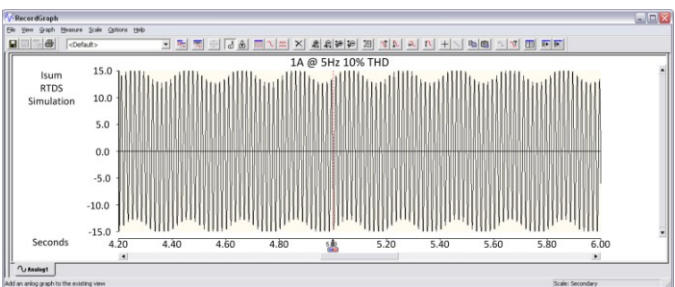


Fig. 10(c): TSHD detector – 5Hz (X axis – time in seconds), (Y axis – current in amps).

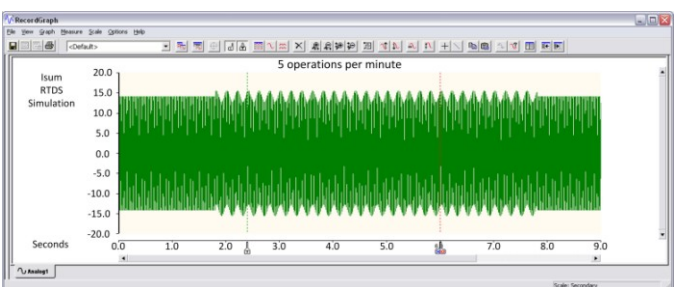


Fig. 10(d): Operation/minute detector – 5 Hz and 5 operations / minute.

In Figure 11, a sequence of events for the operations per minute detector is depicted.

After the installation of the relay at Xcel Energy substation, the relay was tested with the COMTRADE files generated from the actual event to verify the settings.


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2010Sep23 21:55:47.916 Spare 1:OUT1 : Open
2010Sep23 21:55:47.814 Current 1 Detect 1:SHD1: Low
2010Sep23 21:55:47.811 ProLogic 1:PL1 : Low
2010Sep23 21:55:47.811 Current 1 Detect 1:SHD1: High (R)
2010Sep23 21:55:47.811 Spare 1:OUT1 : Closed
2010Sep23 21:55:39.811 ProLogic 1:PL1 : High
2010Sep23 21:55:37.811 ProLogic 1:PL1 : Low
2010Sep23 21:55:29.811 ProLogic 1:PL1 : High
2010Sep23 21:55:27.811 ProLogic 1:PL1 : Low
2010Sep23 21:55:19.811 ProLogic 1:PL1 : High
2010Sep23 21:55:17.811 ProLogic 1:PL1 : Low
2010Sep23 21:55:09.811 ProLogic 1:PL1 : High
2010Sep23 21:55:07.811 ProLogic 1:PL1 : Low
2010Sep23 21:54:59.811 ProLogic 1:PL1 : High

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Fig. 11: Operations/minute event log display, showing the trip after detecting 5 operations.

V. DISCUSSION

Even with simulation studies, sub-harmonic disturbances are difficult to predict. Modern wind turbine systems use advanced electronic devices such as AC/DC convertors which can generate their own harmonics and inter-harmonics. Due to lack of accurate wind generator models from the manufacturers, it is important to capture and analyze the data for protective action.

In cases where commercial technology does not fill the system's need, a successful solution can be found working co-operatively with industry manufacturers. Focusing efforts sharply on solving a very specific real-world technical need was mutually rewarding, and the outcome benefits our industry as a whole.

VI. CONCLUSIONS

The event captured at the Xcel Energy led to the development of a new microprocessor based sub-harmonic protection technique. With the increased use of wind generators feeding HV and EHV utility networks with series compensated lines in the near vicinity, it is necessary to ensure that sub-harmonic oscillations are monitored, and that the electrical grid is protected from any resulting detrimental effects.

VII. REFERENCES

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VIII. BIOGRAPHIES

Krish Narendra obtained his B.E. (Electrical Engineering) in 1986 from University Visweswaraiiah College of Engineering (UVCE), and Msc (E.E), Ph.D. (E.E) with a specialisation in High Voltage Engineering from Indian Institute of Science, India in 1989 and 1993 respectively. He joined the Electrical and Computer Engineering Department of Concordia University, Montreal as a Research Scholar in 1995. From 1996, he worked in software development with APT Technologies and then NxtPhase T&D Corporation, heading Research & Development for the Relay and Recorder division since 2006. Krish served as General Manager of Technology and Quality of ERLPhase Power Technologies Ltd, from June 2007. He is now the VP- Technology and Quality since 2009, and is a member of the core corporate management team.

He has over 18 years of software development expertise and over 25 publications and 3 patents (pending) to his credit. He is actively participating in the IEEE PRSC working groups and is a member of the PRTT of NASPI. His areas of interests include Power Systems Disturbance Analysis, Protection, HVDC Controls, Neural Networks, Fuzzy logic, Phasor Technology (PMUs), and IEC 61850 application to protection and control.



David Fedirchuk graduated from the University of Manitoba with a B Sc degree in Electrical Engineering in 1972. Since then, he has worked for 25 years for Manitoba Hydro, a power utility in the areas of HVDC Transmission, Distribution Design and System Performance. Work in these areas involved commissioning of equipment, study and resolution of power system disturbances and the assessment of operation and application of protective relays within the power system.

In 1985 he became involved in the development of microprocessor based protective relays and recorders and holds several patents for innovation in this area. In 2000, David worked with NxtPhase Corporation as a product development manager to develop product for the power utility industry. David has presented papers to several power system conferences in North America in the areas of power system protection and monitoring. Today, David is working with ERLPhase Power Technologies, in the area of product development helping to develop the next generation of protection and recording technologies.



René Midence (M'2007, SM'2009) For over 25 years, Mr. Midence has been involved in the design and commissioning of power substations and power plants including Protection and Control, SCADA, Substation Automation and Substation LAN systems. His well-rounded experience covers the fields of consulting and engineering, construction and commissioning, manufacturing, strategic marketing, technical support and training.

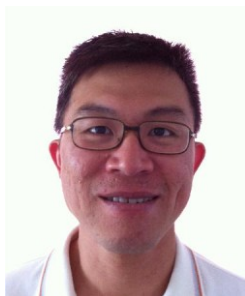
In the manufacturing business, he has worked in the design and testing of medium and high voltage substations as well as metalclad switchgear, protection and control panels. During his more than 15 years of experience in manufacturing, he has contributed to the development and introduction to market of new protection and control microprocessor based relays, Ethernet switches and routers.

Mr. Midence is a Senior Member of the IEEE, have active participation in the development of IEEE Standards, member of the IEEE Power Systems Relaying Committee (PSRC), member of the International Electrotechnical Commission (IEC) TC57 WG10.

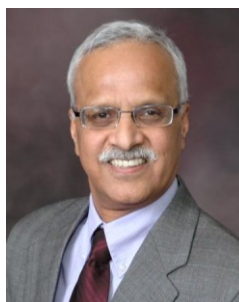
He graduated from the University of Honduras with a Bachelor of Applied Science degree in Electrical and Industrial Engineering.



Nan Zhang obtained his B.E (Electrical Engineering) in 2008 from University of Manitoba. Immediately after his graduation, he joined ERLPhase as a Technical Support EIT in 2008. Besides providing the technical support to the customer, he also involved in the relay algorithm validation and implementation, RTDS simulation testing. He also involved in the product verification including PMU verification and 61850 configuration tool's design and testing.



Adi Mulawarman (SM'11) received his BSEE and MEE degrees in Electrical Engineering from University of Minnesota. He is with Xcel Energy Inc. (NSP) since 1997. He is currently working as consultants sponsor for substation design and engineering department. He is actively involved in the IEEE PES PSRC activities and is the vice-chair of the working group revising Breaker Failure Protection Guide-C37.119-2005.



Pratap Mysore (SM'03) received his BE and ME degrees in Electrical Engineering from Indian Institute of Science, Bangalore, India. He is with HDR Engineering Inc in Minneapolis, MN, USA as National Senior Relay and Protection Engineer. Prior to joining HDR in Jan 2011, he was with Xcel Energy in the substations engineering group since 1987. Pratap is actively involved in the standards development work through IEEE PSRC. He is the chair of Substations subcommittee, chair of the working group revising Shunt capacitor protection guide and member of several working groups within PSRC. With over 30 years of experience in the field of protection and control of power system equipment, Pratap has shared his knowledge and experience with others in the region by presenting papers and offering courses on power system protection at the University of Minnesota. He is one of the recipients of 2011 IEEE Prize paper award and is a recipient of IEEE Twin Cities chapter "outstanding Engineer" award in 2000. He is a registered professional Engineer in the state of Minnesota.



Vijay Sood is an associate professor and the Electrical Engineering Design co-chair at UOIT, Oshawa, Ontario. He has extensive experience in HVDC and FACTS and their controllers. His research focuses on the monitoring, control, and protection of power systems and on the integration of renewable energy systems into the smart grid. He is a registered professional engineer in the province of Ontario. He is a Fellow of the IEEE, the Engineering Institute of Canada and the Canadian Academy of Engineering. He serves as a director of the IEEE Canadian Foundation and is an editor of the IEEE Transactions on Power Delivery, and co-editor of the IEEE Canadian Journal of Electrical and Computer Engineering